MILITARY SPECIFICATION

MINOL-2 COMPOSITION

This Amendment forms a part of Military Specification MIL-M-14745 (MU), dated 15 April 1968.

Paragraph 2.1: Delete "PURCHASE DESCRIPTIONS PA-PD-3087 - Ammonium Nitrate Grained (For Use In Explosives)", and substitute "MIL-A-50460 - Ammonium Nitrate, Prilled (For Use In Ammunition)."

Paragraph 3.1.1: Delete in its entirety and substitute the following:

"3.1.1 Ammonium Nitrate.-The ammonium nitrate shall comply with the requirements of Military Specification MIL-A-50460 (MU). The moisture of the ammonium nitrate shall not exceed 0.15 percent when tested as specified in 4.2.2."

Paragraph 3.2 - Composition, Table I: Delete the following:

"Percent
40 ± 3
20 ± 2
40 ± 3 "

And substitute:

"Percent
40 ± 3
20 ± 3
40 ± 3 "

Paragraph 3.5 : Delete in its entirety.

FSC: 1376

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Paragraph 4.2.3, Sampling for Test: Delete the following: "Samples from kettle from each line" and substitute: "The samples shall be taken from the discharge, spout (leg) of the main pouring kettle immediately before entering the bombs at each line and shift."

Paragraph 4.3.1: Delete in its entirety and substitute the following:

"4.3.1 Determination of charge composition.-The specimen shall be taken from a sample approximately four (4) inches in diameter with a maximum thickness of one-quarter (1/4) inch that has been ground to pass a number 10 mesh sieve. All weighing shall be done on an analytical balance to plus or minus 0.0002 gram. The specimen selected from the ground sample shall be approximately 2 grams, accurately weighed, and shall be placed in a tared, medium-porosity, sintered glass crucible. A stirring rod and about 20 ml of benzene shall be added to the crucible and the contents shall be stirred three times for about 30 seconds each time. The crucible and contents shall be placed on a filtration apparatus and suction applied to draw out the benzene solution. With the solution still on, the walls of the crucible and the stirring rod shall be washed down with additional benzene. The entire procedure, starting with the addition of 20 ml of benzene, shall be repeated two additional times in order to insure complete extraction of the TNT. After the third extraction, remove the stirring rod after rinsing with benzene to insure that all adhering particles are washed into the crucible and continue the suction for about a minute after all the benzene solution has been drawn through. The crucible, which now contains ammonium nitrate and aluminum, shall be placed in an oven at 100 degrees plus or minus 5 degrees C for approximately 15 minutes, cooled in a desiccator, and weighed.

After weighing the stirring rod shall be replaced in the crucible and about 20 ml of distilled water shall be added. As before, the contents shall be stirred three times, for about 30 seconds each time, the solution shall be drawn through the crucible by applying suction and the entire process repeated for a total of three extractions with 20 ml of water each time. After the third extraction, the stirring rod shall be washed with water and removed, as before, but the final water wash shall be followed by an acetone rinse in order to displace the water and facilitate drying. After the acetone rinse, the crucible shall be aspirated for about a minute before transferring to the oven. The crucible, which now contains only aluminum, shall be dried for approximately 15 minutes at 100 degrees plus or minus 5 degrees C, cooled in a desiccator, and weighed. The composition shall be calculated as follows:

$$\text{TNT } \% = \frac{W_1 - W_2}{0.01\ W}$$

$$\text{Ammonium Nitrate } NH_4NO_3\ \% = \frac{W_2 - W_3}{0.01\ W}$$

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$$\text{Aluminum \%} = \frac{W_3 - W_0}{0.01W}$$

WHERE:

W = Weight of specimen, gram
W_0 = Weight of empty crucible, gram
W_1 = W + W_0, gram
W_2 = Weight of crucible, $NH_4 NO_3$ and Al, gram
W_3 = Weight of crucible and Al, gram

CUSTODIAN:
ARMY-MU

PREPARING ACTIVITY:
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